

- 8 A pendulum bob of mass 1.27 kg is supported by a string so that the radius of its path is 0.600 m. It is moving with velocity 0.575 m s^{-1} horizontally at the centre of its motion when the string is vertical.

What is the tension in the string at this instant?

A 11.8 N

B 12.5 N

C 13.2 N

D 13.7 N

C $T - mg = mv^2/r$

$$T = m(g + v^2/r)$$

$$T = 1.27(9.81 + 0.575^2/0.600) = 13.2 \text{ N}$$

